Lecture 9

2/9/26

Homework this week is mostly "hands on."

- **Activity 4.1** Resolution & spatial filtering

Use diffraction gratings to look at effects of line density, orientation, etc. on diffraction pattern

Location: laser room in back of Olin 237

- **Activity 4.2** Kohler illumination, Abbe theory, ...

Use horizontal μscope on breadboard to study Kohler illumination, conjugate planes, Abbe theory, etc.

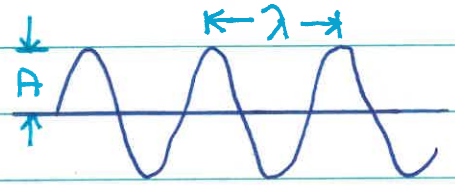
Location: Olin 237

A4.1 – do own your own
A4.2 – sign up for time slot
} due in ~2 wks

Now getting into more sophisticated aspects of imaging

Know minimum resolvable spacings (e.g., from Abbe)

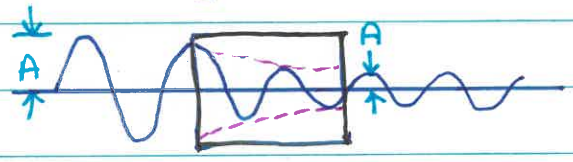
Eye is sensitive to two aspects of light waves



- A = amplitude (brightness)
- λ = wavelength (color)

Brightness variations can have various specimen-dependent origins:

- Specimen differentially absorbs, blocks, deflects light

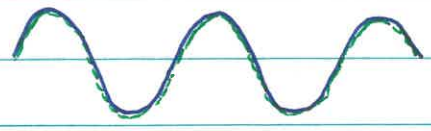


A is reduced by absorption w/in sample feature

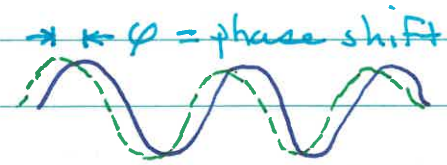
- Specimen differentially binds fluorophores (fluorescent labels) or stains (colored dyes)

Most biological specimens (e.g., cells) are intrinsically transparent: they <u>don't</u> alter amplitude
they <u>don't</u> have color variations

Most biological specimens <u>do</u> alter wave phase (i.e., the location of peaks in the waves)



Waves in Phase



Waves Out of Phase

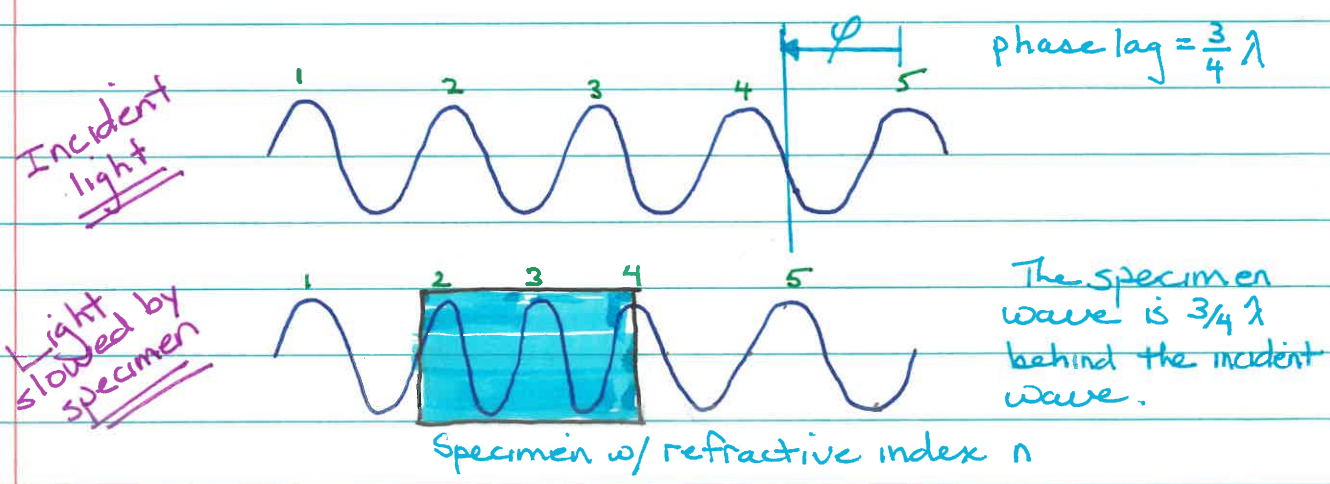
Light travels more slowly in media w/ higher refractive index

Slowed speed $v = \frac{c}{n}$

c ← speed of light in vacuum

n ← refractive index of medium

Phase shifts occur w/ specimens because the higher refractive index of the specimen slows light relative to air and causes the waves to be out of sync



The eye can't detect the phase shift, which is why cells are normally transparent

What can be done?

Brightfield Microscopy (BF)

BF uses standard Kohler illumination

BF is used to study absorbing (amplitude) specimens. Histology, for example, involves using BF to image fixed (dead) stained cells. Big for cancer diagnosis & bacterial identification

Beyond Brightfield

There are three common "tricks" to enhance contrast w/ phase specimens

(1) Alter illumination

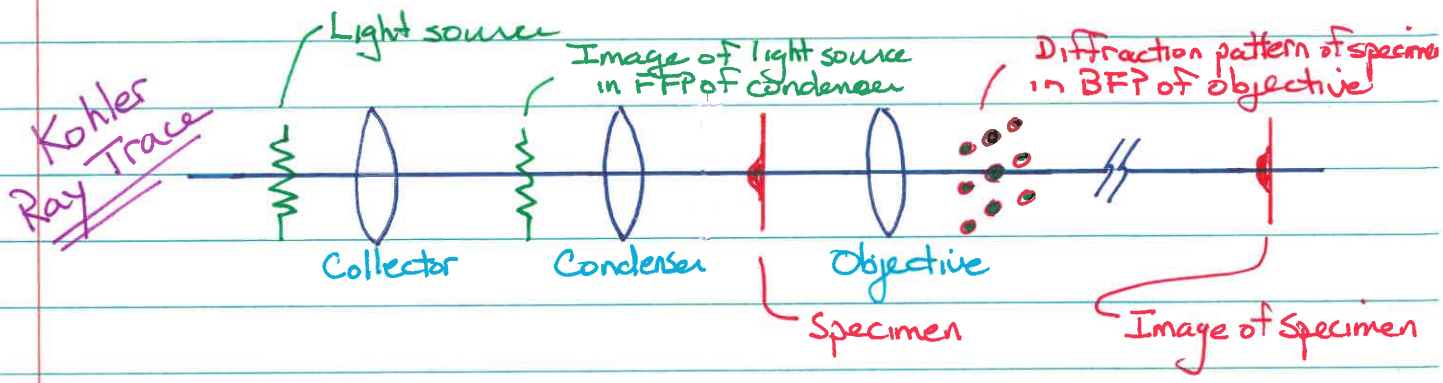
Image of light source in FFP of condenser lens

(2) Filter diffraction pattern from specimen

Specimen diffraction pattern in BFP of objective lens

(3) Use interference

Use interference between illumination light & specimen-affected light to brighten or dim portions of image in specimen-dependent way



Key points for contrast generation:

- Can manipulate illumination in FFP of condenser
- Can manipulate specimen diffraction pattern in BFP of objective

ACTIVITY 4 Kohler Illumination

Walked thru Activity together, w/ Bethe leading, & emphasis on questions 1 & 3

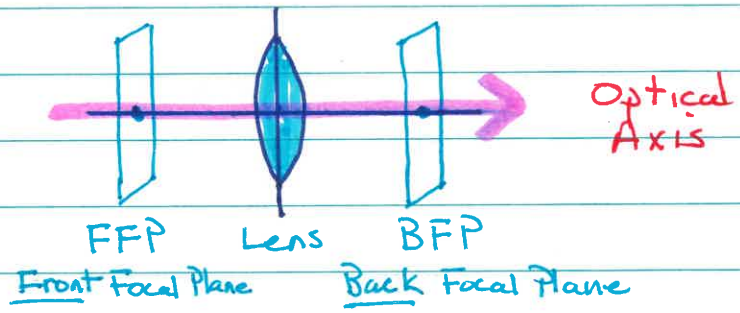
MEMORIZE Kohler diagram, including μscope components & places (conjugate planes) where specimen & light source are in focus

Quiz will ask students to fill out Activity 4 from memory in a future class

Kohler Illumination

Need to understand the μscope optical path to understand how these contrast-enhancing tricks are implemented

Looked at Kohler illumination ray trace < Power Point / Text - Fig 3.13



There are two primary lenses in the optical μscope, positioned just before & just after the specimen

(1) Condenser Lens – involved in illumination
- Impt in setting up Kohler illumination
- Light source is in focus in FFP of condenser

(2) Objective Lens – involved in image formation
- Collects light from specimen
- Specimen diffraction pattern is in BFP of objective

Darkfield Microscopy (DF)

DF is an application of contrast enhancement

DF is the opposite of BF:

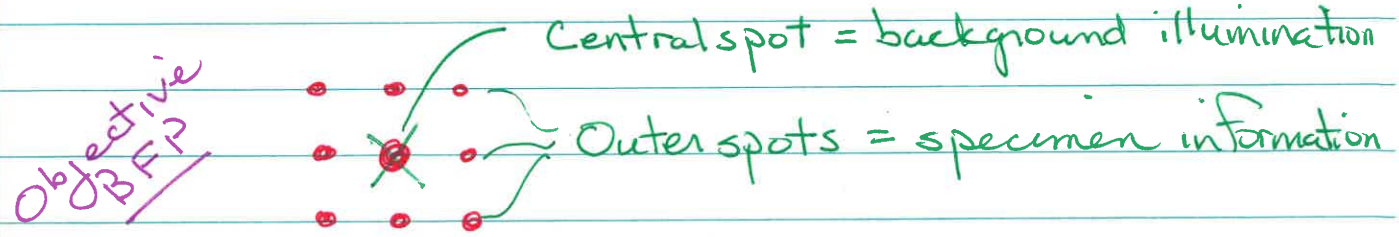
- BF – bright background
  see specimen as dark against bright
- DF – dark background
  see specimen as bright against dark

Analogy – stars are out all the time
can't see stars against bright daytime sky
can see stars against dark nighttime sky

It's easier to see bright things against a dark background.

There are two methods for implementing DF:

(1) Filter diffraction pattern in objective BFP



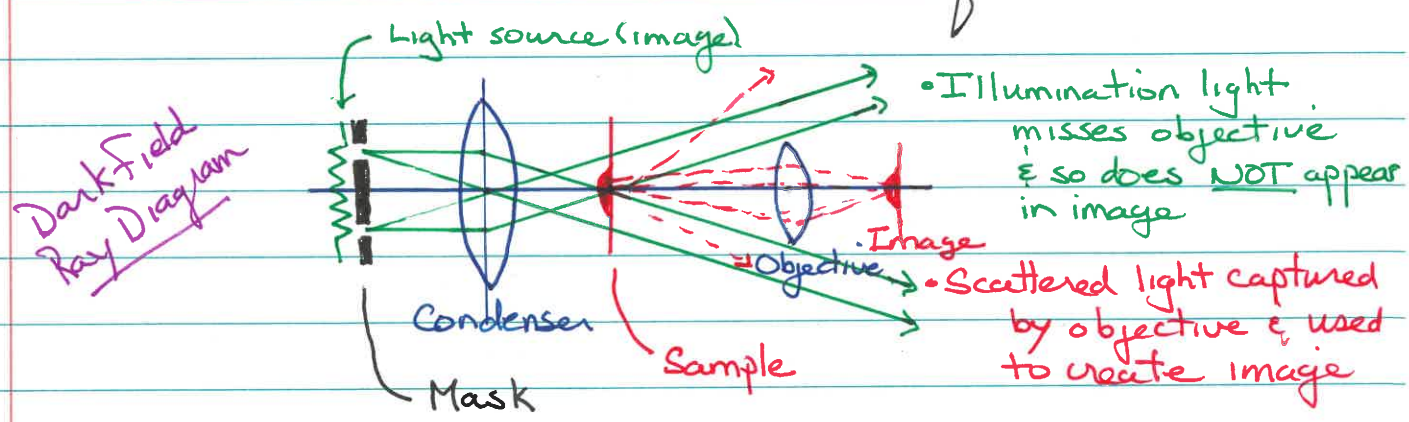
Remove central spot ⇒ Background field will go dark
Removes illumination that is <u>NOT</u> affected by the specimen

The illumination in DF is uniform bright Kohler background

(1) <u>Alter illumination</u> in FFP of condenser

It's often difficult to access the BFP of the objective, so normally implement DF a different way

Alter illumination in the FFP of the condenser



Put mask in FFP of condenser where image of light source appears.
Only angled illumination light gets thru opening(s) in mask.
Angled light misses objective & so does <u>NOT</u> contribute to image.
Only diffracted light is captured by objective to form image